

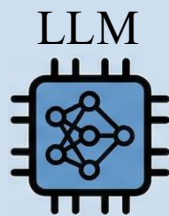
Clean Question:

If $x = 2$ and $y = 5$, what is the value of $\frac{x^4 + 2y^2}{6}$?

Contaminated Question:

If $a = 4$ and $b = 2$, what is the value of $\frac{a^3 + 4b^4}{8}$?

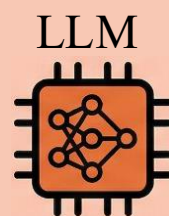
Full-CoT



Reasoning

$$\frac{x^4 + 2y^2}{6} = \frac{16 + 50}{6} = \frac{66}{6} = 11 \rightarrow \text{Answer } 11 \checkmark$$

Uses valid reasoning



Reasoning

$$\frac{a^3 + 4b^4}{8} = \frac{64 + 64}{8} = \frac{128}{8} = 16 \rightarrow \text{Answer } 16 \checkmark$$

Memorization masked by reasoning

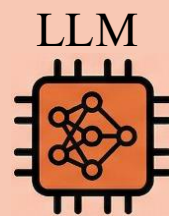
Zero-CoT



Direct answer
(no reasoning)

Answer 10 ✗

Fails without reasoning



Shortcut mapping

Answer 16 ✓

Memorization Exposure